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FIB Install Rate in PTX Express5

An HPE Juniper Networking Techpost

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1 FIB Install Rate in PTX Express5

Vivek Singh Sikarwar - 05/31/2024

PTX10002-36QDD is the first router equipped with the new Juniper Express5 packet forwarding engine, a new deep-buffer 28.8Tbps package introducing a lot of innovations and improvements compared to its predecessor.

This post will answer a simple question: how fast can we program the FIB entries in hardware on this new chipset?

TL;DR: that's around 38,000 prefixes / second.

Tests prepared, executed and originally documented by Vivek Singh Sikarwar. Article written by Nicolas Fevrier.

1.1 Introduction

The new Juniper Networks Packet Forwarding Engine (PFE) is now available in shipping products, for instance the PTX10002-36QDD. We will use this new router for the test and will follow the same methodology described in this article: https://techpost.hpe.com/articles/ptx10001-36mr-fib-install-rate/article

1.2 Test Methodology

It consists in using an intermediate helper / shadow router then:

- inject the full routing table from the Ixia testing device
- generate traffic for all prefixes
- enable the links between the Device Under Test (DUT) and the helper router
- measure how long it takes to have Tx traffic equals Rx traffic

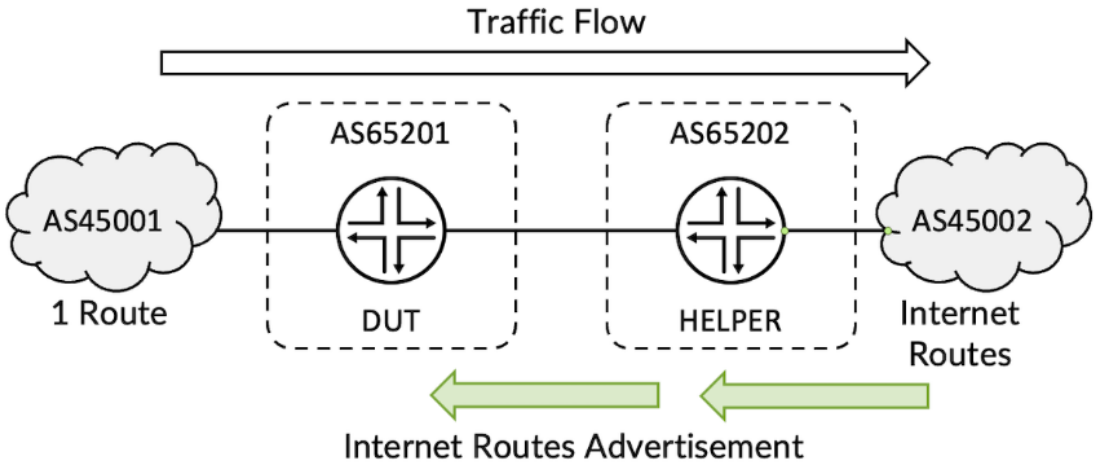


Figure 1: Test topology

The packet size used for this test is following this distribution:

Packet Size (Bytes)	Weight IPv4 IMIX	Weight IPv6 IMIX
64	3	0
78	0	3
100	26	26
373	6	6
570	5	5
1300	6	6
1518	16	16
9000	1	1

EBGP peering is enabled across the routers as well as across the router and traffic generator. The DUT learns a single route from the traffic generator and 970k IPv4 Internet prefixes from the helper DUT. Traffic is sent from the single route as the source and internet routes as the destination with 99.9% of offer load in iMIX mode.

The test starts with disabling the EBGP peering between the DUT and helper DUT, such that traffic drops and Tx is MAX rate and Rx is ZERO.

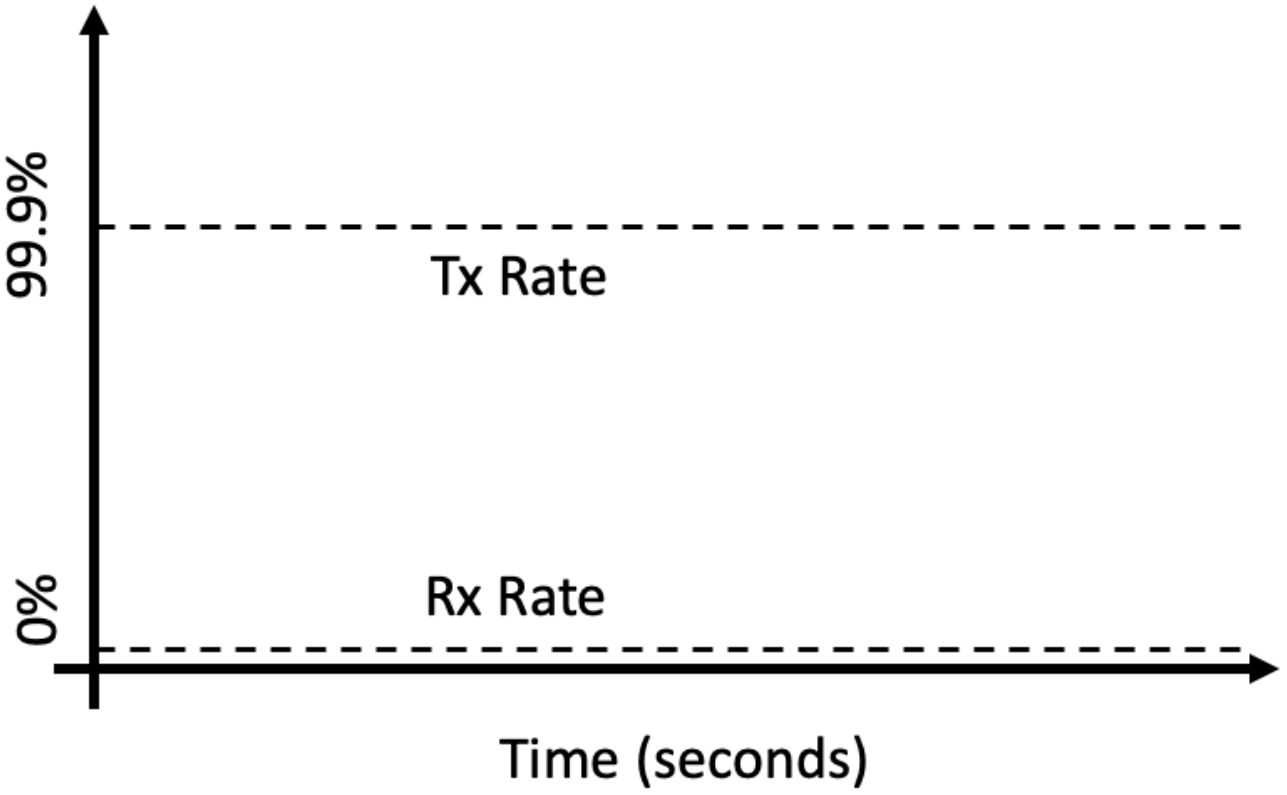


Figure 2: Graph 1

Once the interface between the routers is enabled, BGP peering will establish and Helper DUT advertise the internet prefixes to the DUT and Rx Rate increases as DUT installs the routes.

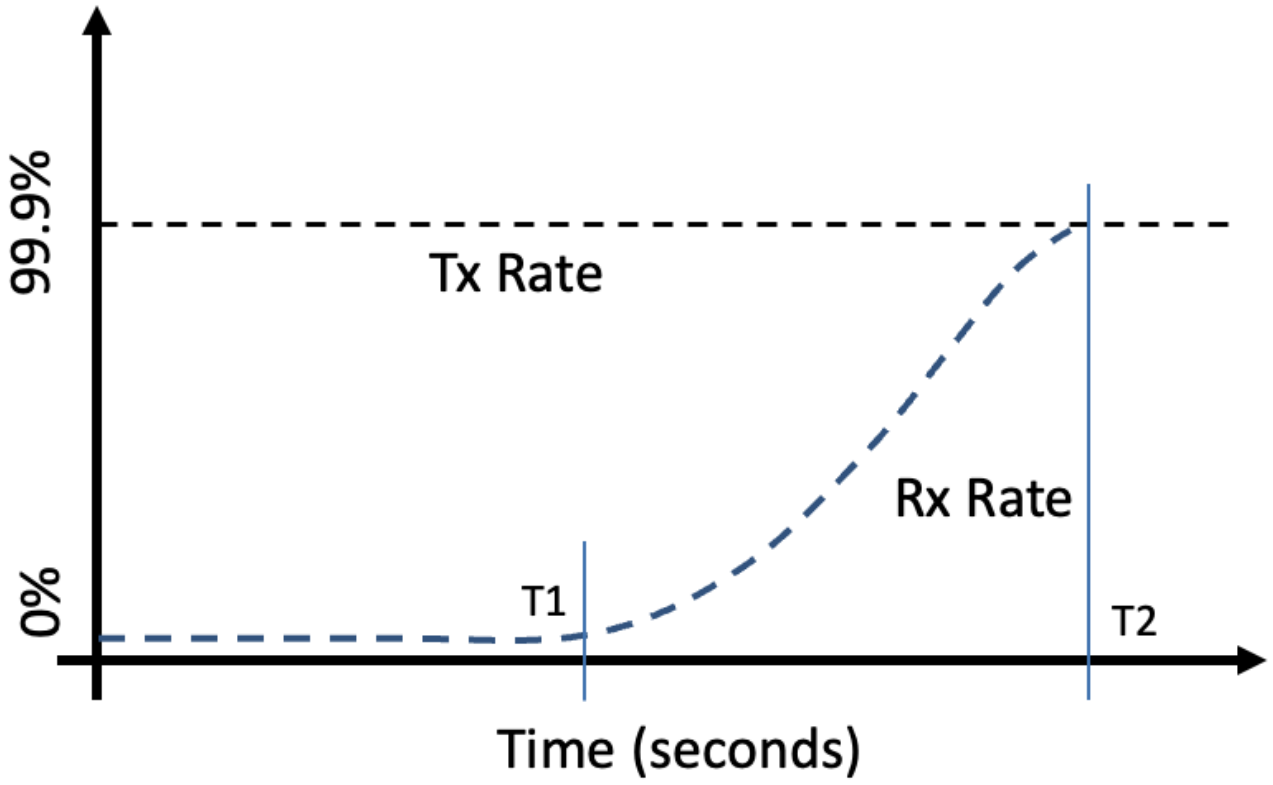


Figure 3: Graph 2

The total routes divided by T2-T1 gives the number of routes installed per second.

1.3 Test Results

Each test is executed three times for IPv4 and three times for IPv6. We found consistent results for each test category.

1.3.1 Test Results with IPv4 Internet Profile



Figure 4: Ixia Configuration for IPv4 Routes

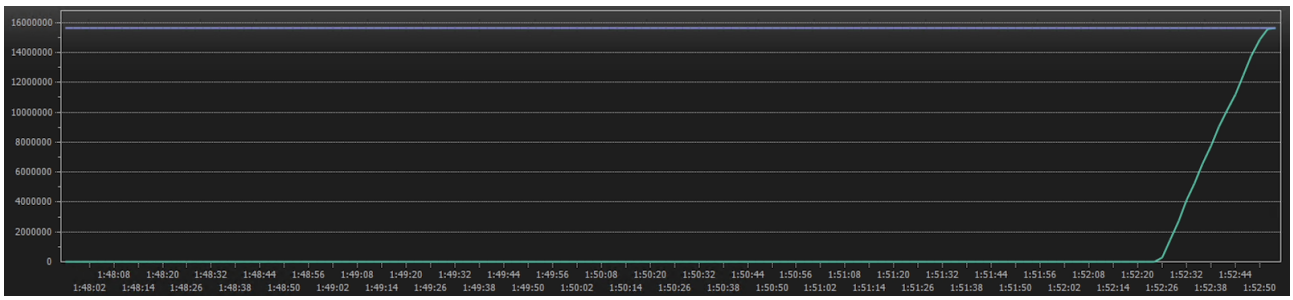
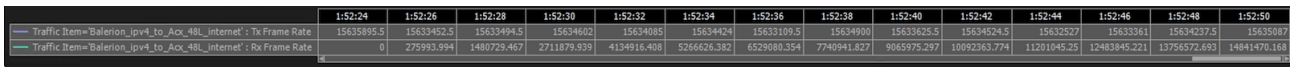


Figure 5: Rx/Tx Traffic Graph for IPv4 Routes



	1:52:24	1:52:26	1:52:28	1:52:30	1:52:32	1:52:34	1:52:36	1:52:38	1:52:40	1:52:42	1:52:44	1:52:46	1:52:48	1:52:50
Traffic Item='Balerion_ipv4_to_Acx_48L_internet' : Tx Frame Rate	15635895.5	15633452.5	15633494.5	15634602	15634085	15634424	15633109.5	15634900	15633625.5	15634524.5	15632527	15633361	15634237.5	15635087
Traffic Item='Balerion_ipv4_to_Acx_48L_internet' : Rx Frame Rate	0	275993.994	1480729.467	2711879.939	4134916.408	5266626.382	6529080.354	7740941.827	9065975.297	10092363.774	11201045.25	12483845.221	13756572.693	14841470.168

Figure 6: Rx/Tx Traffic Graph for IPv4 Routes

Despite the large routing table, the routes are programmed very fast in the system and the traffic/route generator only displays information with a cadence of 2 seconds.

Let's try to calculate an average of the programming speed during this route advertisement sequence.

We generate 15,633,625 PPS on 970,304 routes. That represents 16,1121228 PPS for each route in average.

The graph above starts between 01:52:26 and 01:52:28 and Rx=Tx between 01:52:50 and 01:52:52, so we will skip these two intervals and focus our attention on the intervals between these two points.

Time	PPS Received	Delta of PPS Received(2s interval)	Install Rate in pfx/sfor this 2s Interval
01:52:26	275,993	-	-
01:52:28	1,480,729	602,368	37,386
01:52:30	2,711,879	615,575	38,206
01:52:32	4,134,916	711,518.5	44,160
01:52:34	5,266,626	565,855	35,120
01:52:36	6,529,080	631,227	39,177
01:52:38	7,710,941	590,930.5	36,676
01:52:40	9,065,974	677,517	42,050
01:52:42	10,092,363	513,194	31,851
01:52:44	11,201,045	554,341	34,405
01:52:46	12,483,845	641,400	39,808
01:52:48	13,756,572	636,363	39,495
01:52:50	14,841,470	542,449	33,667

In average, during this installation period, we installed 37,667 IPv4 prefixes per second.

1.3.2 Test Results with IPv6 Internet Profile

Same approach will be used for IPv6.

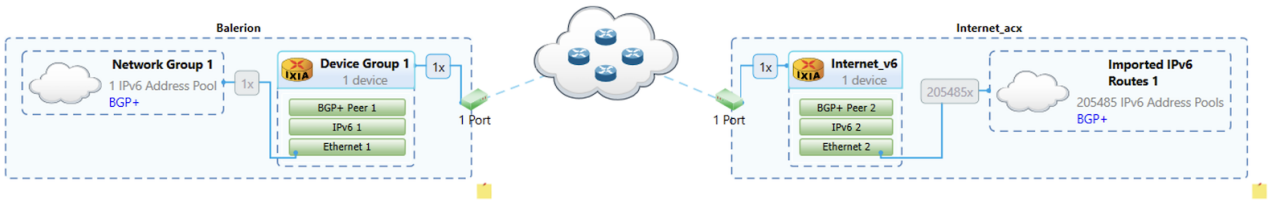


Figure 7: Ixia Configuration for IPv6 Routes

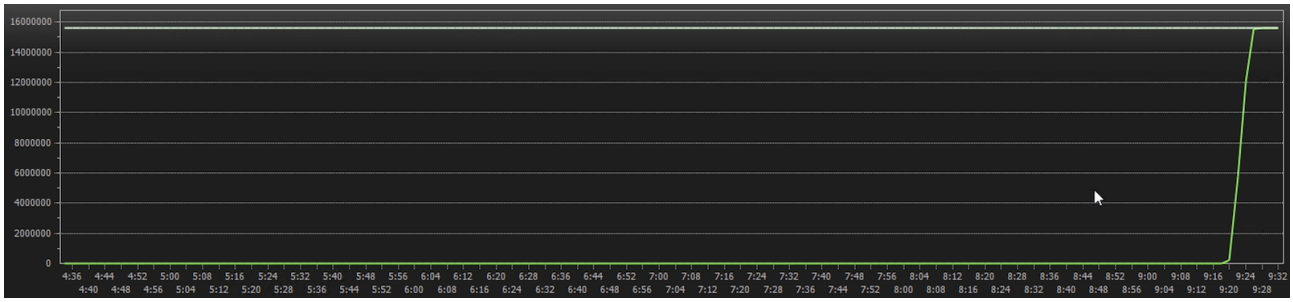


Figure 8: Rx/Tx Traffic Graph for IPv6 Routes

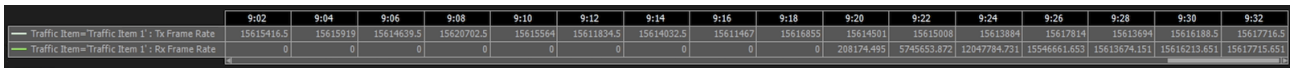


Figure 9: Rx/Tx Traffic Graph for IPv6 Routes

We generate 15,616,855 PPS on 205,485 routes. That represents 76 PPS for each route in average.

The graph above shows that traffic starts being received between 9:20 and 9:22 and Rx=Tx between 9:26 and 9:28. If doesn't give us much intervals to really monitor the installation rate.

Time	PPS Received	Delta of PPS Received(2s interval)	Install Rate in pfx/sfor this 2s Interval
9:20	208,174	-	-
9:22	5,745,653	2,768,739.5	36,431
9:24	12,047,784	3,151,065.5	41,461
9:26	15,546,661	1,749,438.5	23,019
9:28	15,617,814	-	-

That gives us only three datapoints to draw our average, which is not a lot from a statistical point of view. Nevertheless, this average is 33,637 prefixes per second.

1.4 Conclusion

Installation rate has been optimised on the Express5 powering the PTX10002-36QDD. It exceeds 37,500 prefixes per second. With an internet table reaching the million IPv4 entry in mid-2024, it takes less than 27 second to complete the programming in hardware.

1.5 Useful links

- Express 5 Overview: https://techpost.hpe.com/articles/express-5-overview/article
- Introducing PTX10002-36QDD: https://techpost.hpe.com/articles/introducing-ptx10002-36qdd/article
- FIB Scale in Express5: https://techpost.hpe.com/articles/fib-scale-in-express5/article
- Tests on PTX10001-36MR: https://techpost.hpe.com/articles/ptx10001-36mr-fib-install-rate/article

1.6 Glossary

- DUT: Device Under Test
- EBGP: External Border Gateway Protocol
- FIB: Forwarding Information Base
- PFE: Packet Forwarding Engine
- PPS: Packets per Second

1.7 Acknowledgments

Tests prepared, executed and originally documented by Vivek Singh Sikarwar. Article written by Nicolas Fevrier. Thanks to Suneesh Babu and Anson Varghese Isaacs for original tests and support.



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